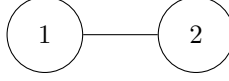


Time Needed for One-Step PST

Nathan Ngo
(Dated: October 2024)

I. P_2



A. Analytical

The graph is represented by the Hamiltonian

$$H = -J \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (1)$$

This Hamiltonian indicates that the probability amplitude can "hop" between the two nodes. We then need to find the eigenvalues of H :

$$\begin{aligned} \begin{vmatrix} -\lambda & -J \\ -J & -\lambda \end{vmatrix} &= 0 \\ \lambda^2 - J^2 &= 0 \\ \lambda_1 = J & \quad \lambda_2 = -J \end{aligned} \quad (2)$$

and their eigenvectors are

$$u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (3)$$

The time evolution of the quantum state is given by

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle \quad (4)$$

We know that H can be decomposed to

$$H = \lambda_1 |u_1\rangle\langle u_1| + \lambda_2 |u_2\rangle\langle u_2| \quad (5)$$

By the approximation $e^x = \sum_{n=0}^{\infty} x^n/n!$, we have

$$e^{-iHt/\hbar} = e^{-i\lambda_1 t/\hbar} |u_1\rangle\langle u_1| + e^{-i\lambda_2 t/\hbar} |u_2\rangle\langle u_2| \quad (6)$$

The proof of Eq. 6 can be found in the appendix. To solve for t in Eq. 4, we still need $|\psi(0)\rangle$ and $|\psi(t)\rangle$. Assuming that the initial state is localised on node 1, we have

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (7)$$

For a perfect quantum state transfer, the final state should be localised on node 2, so

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (8)$$

Putting Eq. 6, Eq. 7 and Eq. 8 into Eq. 4, we have

$$\begin{aligned}
\begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \left(e^{-i\lambda_1 t/\hbar} |u_1\rangle \langle u_1| + e^{-i\lambda_2 t/\hbar} |u_2\rangle \langle u_2| \right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \left[e^{-iJt/\hbar} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} + e^{iJt/\hbar} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \right] \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \left[e^{-iJt/\hbar} \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} + e^{iJt/\hbar} \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix} \right] \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} e^{-iJt/\hbar} + e^{iJt/\hbar} & e^{-iJt/\hbar} - e^{iJt/\hbar} \\ e^{-iJt/\hbar} - e^{iJt/\hbar} & e^{-iJt/\hbar} + e^{iJt/\hbar} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} e^{-iJt/\hbar} + e^{iJt/\hbar} \\ e^{-iJt/\hbar} - e^{iJt/\hbar} \end{pmatrix} \\
\begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} \cos \frac{Jt}{\hbar} \\ -i \sin \frac{Jt}{\hbar} \end{pmatrix} \\
t &= \frac{1}{2} \frac{\pi \hbar}{J}
\end{aligned} \tag{9}$$

Therefore, the quantum state at node 1 is transferred to node 2 with perfect fidelity after $\frac{\pi \hbar}{2J}$ seconds. To make it dimensionless, we can define the dimensionless time as

$$t = a \tilde{t} \tag{10}$$

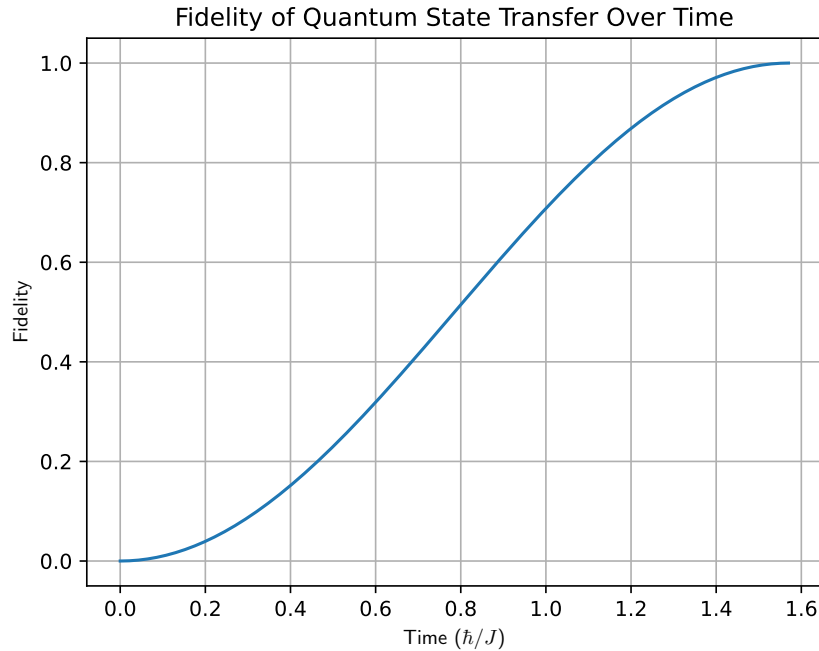
where

$$a = \frac{\hbar}{J} \tag{11}$$

The dimensionless time in this case is

$$\tilde{t} = \frac{t}{a} = \frac{\pi \hbar}{2J} \frac{J}{\hbar} = \frac{\pi}{2} \tag{12}$$

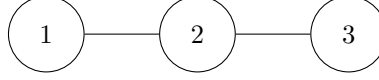
B. Numerical



The dimensionless time calculated is

$$\tilde{t} = 1.5698 \quad (13)$$

II. P_3



A. Analytical

The graph is represented by the Hamiltonian

$$H = -J \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad (14)$$

We then need to find the eigenvalues of H :

$$\begin{aligned} & \begin{vmatrix} -\lambda & -J & 0 \\ -J & -\lambda & -J \\ 0 & -J & -\lambda \end{vmatrix} = 0 \\ & (-\lambda)(\lambda^2 - J^2) + J^2 = 0 \\ & -\lambda^3 + 2\lambda J^2 = 0 \\ & \lambda(\lambda^2 - 2J^2) = 0 \\ & \lambda_1 = 0 \quad \lambda = \sqrt{2}J \quad \lambda_3 = -\sqrt{2}J \end{aligned} \quad (15)$$

and their eigenvectors are

$$u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad u_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad u_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (16)$$

Assuming the initial state is localised on node 1, we have

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (17)$$

For a perfect quantum state transfer, the final state should be localised on node 2, so

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (18)$$

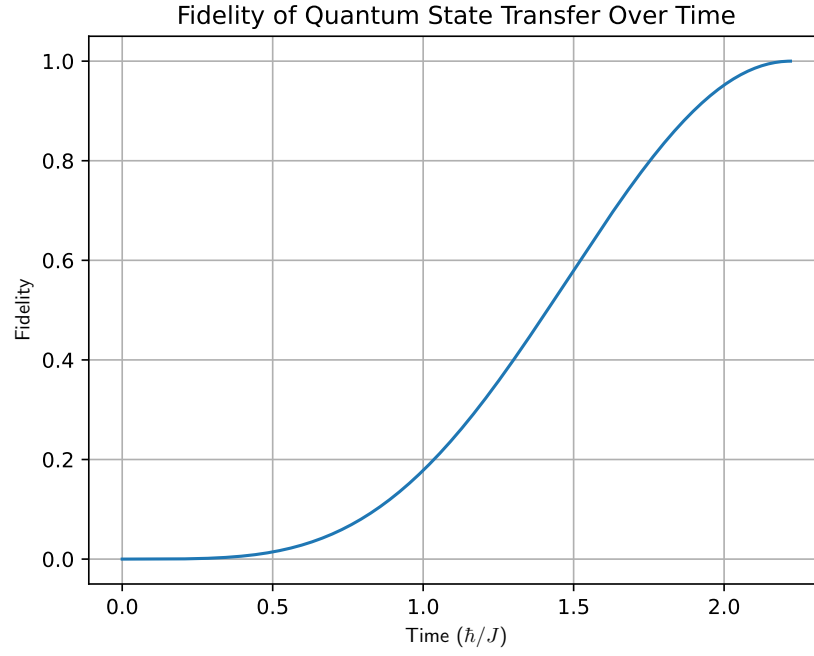
Now, we have

$$\begin{aligned}
\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} &= \left(e^{-i\lambda_1 t/\hbar} |u_1\rangle \langle u_1| + e^{-i\lambda_2 t/\hbar} |u_2\rangle \langle u_2| + e^{-i\lambda_3 t/\hbar} |u_3\rangle \langle u_3| \right) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} &= \left[\begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix} (1/\sqrt{2} \ 0 \ -1/\sqrt{2}) + e^{-i\sqrt{2}Jt/\hbar} \begin{pmatrix} 1/2 \\ 1/\sqrt{2} \\ 1/2 \end{pmatrix} (1/2 \ 1/\sqrt{2} \ 1/2) + e^{i\sqrt{2}Jt/\hbar} \begin{pmatrix} 1/2 \\ -1/\sqrt{2} \\ 1/2 \end{pmatrix} (1/2 \ -1/\sqrt{2} \ 1/2) \right] \\
\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} &= \left[\begin{pmatrix} 1/2 & 0 & -1/2 \\ 0 & 0 & 0 \\ -1/2 & 0 & 1/2 \end{pmatrix} + e^{-i\sqrt{2}Jt/\hbar} \begin{pmatrix} 1/4 & 1/2\sqrt{2} & 1/4 \\ 1/2\sqrt{2} & 1/2 & 1/2\sqrt{2} \\ 1/4 & 1/2\sqrt{2} & 1/4 \end{pmatrix} + e^{i\sqrt{2}Jt/\hbar} \begin{pmatrix} 1/4 & -1/2\sqrt{2} & 1/4 \\ -1/2\sqrt{2} & 1/2 & -1/2\sqrt{2} \\ 1/4 & -1/2\sqrt{2} & 1/4 \end{pmatrix} \right] \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} \left(1/2 + e^{-i\sqrt{2}Jt/\hbar}/4 + e^{i\sqrt{2}Jt/\hbar}/4 \right) & \left(e^{-i\sqrt{2}Jt/\hbar}/2\sqrt{2} - e^{i\sqrt{2}Jt/\hbar}/2\sqrt{2} \right) & \left(-1/2 + e^{-i\sqrt{2}Jt/\hbar}/4 + e^{i\sqrt{2}Jt/\hbar}/4 \right) \\ \left(e^{-i\sqrt{2}Jt/\hbar}/2\sqrt{2} - e^{i\sqrt{2}Jt/\hbar}/2\sqrt{2} \right) & \left(e^{-i\sqrt{2}Jt/\hbar}/2 + e^{i\sqrt{2}Jt/\hbar}/2 \right) & \left(e^{-i\sqrt{2}Jt/\hbar}/2\sqrt{2} - e^{i\sqrt{2}Jt/\hbar}/2\sqrt{2} \right) \\ \left(-1/2 + e^{-i\sqrt{2}Jt/\hbar}/4 + e^{i\sqrt{2}Jt/\hbar}/4 \right) & \left(e^{-i\sqrt{2}Jt/\hbar}/2\sqrt{2} - e^{i\sqrt{2}Jt/\hbar}/2\sqrt{2} \right) & \left(1/2 + e^{-i\sqrt{2}Jt/\hbar}/4 + e^{i\sqrt{2}Jt/\hbar}/4 \right) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} 1/2 + e^{-i\sqrt{2}Jt/\hbar}/4 + e^{i\sqrt{2}Jt/\hbar}/4 \\ e^{-i\sqrt{2}Jt/\hbar}/2\sqrt{2} - e^{i\sqrt{2}Jt/\hbar}/2\sqrt{2} \\ -1/2 + e^{-i\sqrt{2}Jt/\hbar}/4 + e^{i\sqrt{2}Jt/\hbar}/4 \end{pmatrix} \\
\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} \left(1 + \cos \left(\frac{\sqrt{2}Jt}{\hbar} \right) \right) \\ -\frac{i}{\sqrt{2}} \sin \left(\frac{\sqrt{2}Jt}{\hbar} \right) \\ \frac{1}{2} \left(-1 + \cos \left(\frac{\sqrt{2}Jt}{\hbar} \right) \right) \end{pmatrix} \\
t &= \frac{\pi\hbar}{\sqrt{2}J}
\end{aligned} \tag{19}$$

or in dimensionless terms,

$$\tilde{t} = \frac{\pi}{\sqrt{2}} \tag{20}$$

B. Numerical

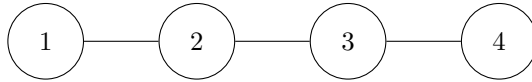


The dimensionless calculated is

$$\tilde{t} = 2.2205 \quad (21)$$

which agrees with the analytical result.

III. P_4



A. Analytical

The graph is represented by the Hamiltonian

$$H = -J \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (22)$$

Using Mathematica, the eigenvalues are found to be

$$\lambda_1 = \frac{J}{2}(-1 - \sqrt{5}) \quad \lambda_2 = \frac{J}{2}(1 + \sqrt{5}) \quad \lambda_3 = \frac{J}{2}(1 - \sqrt{5}) \quad \lambda_4 = \frac{J}{2}(-1 + \sqrt{5}) \quad (23)$$

and their eigenvectors are

$$u_1 = \begin{pmatrix} 1 \\ \frac{1}{2}(1 + \sqrt{5}) \\ \frac{1}{2}(1 + \sqrt{5}) \\ 1 \end{pmatrix} \quad u_2 = \begin{pmatrix} -1 \\ \frac{1}{2}(1 + \sqrt{5}) \\ \frac{1}{2}(-1 - \sqrt{5}) \\ 1 \end{pmatrix} \quad u_3 = \begin{pmatrix} -1 \\ \frac{1}{2}(1 - \sqrt{5}) \\ \frac{1}{2}(-1 + \sqrt{5}) \\ 1 \end{pmatrix} \quad u_4 = \begin{pmatrix} 1 \\ \frac{1}{2}(1 - \sqrt{5}) \\ \frac{1}{2}(1 - \sqrt{5}) \\ 1 \end{pmatrix} \quad (24)$$

B. Numerical

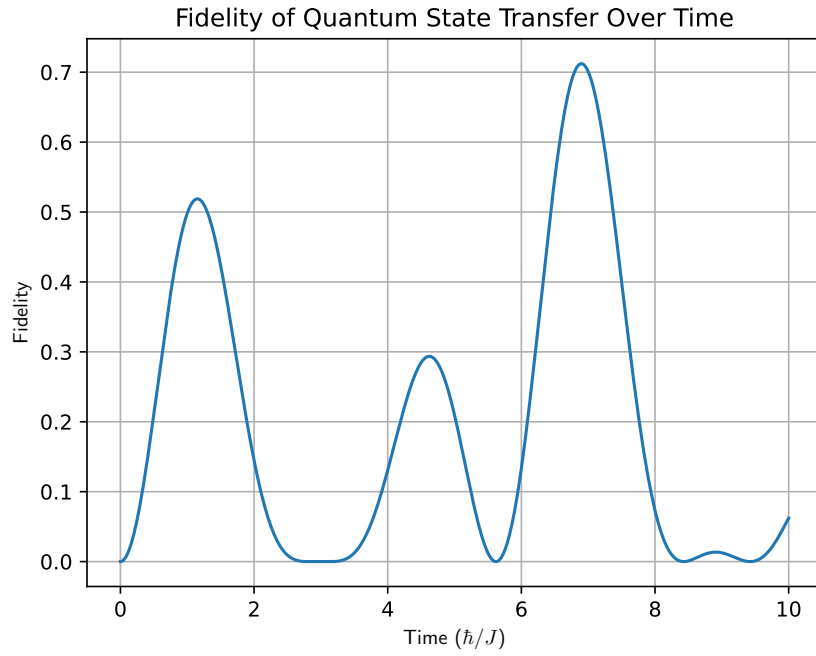
Assuming the initial state is localised on node 1, we have

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (25)$$

I am not sure about what the target state should be, so I tried different cases.

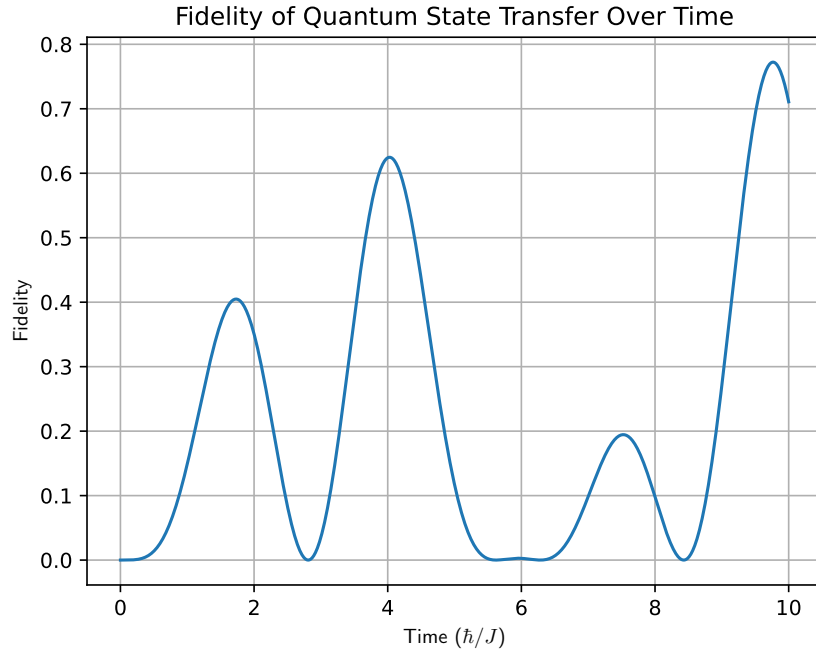
- When the target state is

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (26)$$



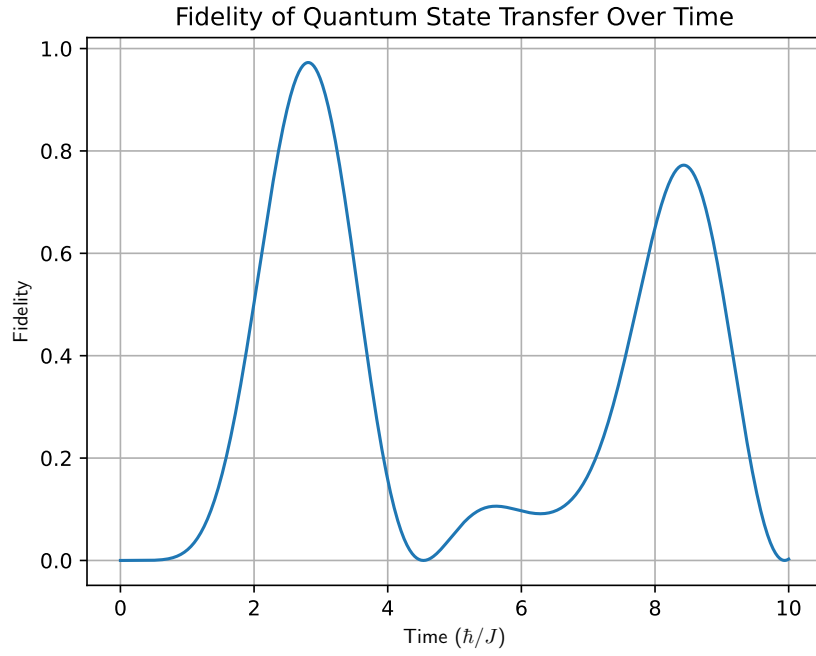
- When the target state is

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad (27)$$



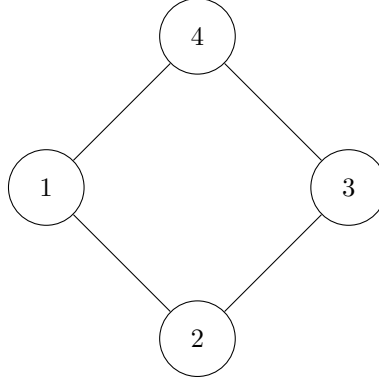
- When the target state is

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (28)$$



We can see that in the first 10 seconds, although a high-fidelity transfer can take place from node 1 to node 4, it is not a perfect quantum state transfer.

IV. C_4



A. Analytical

The graph is represented by the Hamiltonian

$$H = -J \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \quad (29)$$

Using Mathematica, the eigenvalues are found to be

$$\lambda_1 = -J\sqrt{3} \quad \lambda_2 = J\sqrt{3} \quad \lambda_3 = 0 \quad \lambda_4 = 0 \quad (30)$$

and their eigenvectors are

$$u_1 = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ 1 \\ \frac{2}{\sqrt{3}} \\ 1 \end{pmatrix} \quad u_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ 1 \\ -\frac{2}{\sqrt{3}} \\ 1 \end{pmatrix} \quad u_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad u_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (31)$$

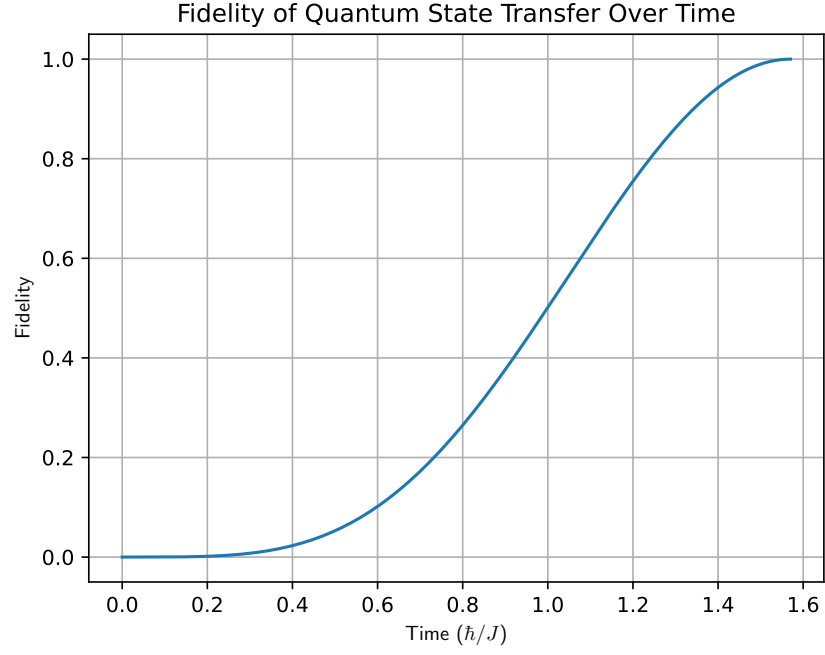
B. Numerical

Assuming the initial state is localised on node 1, we have

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (32)$$

By symmetry, the final state should be localised on node 3, so

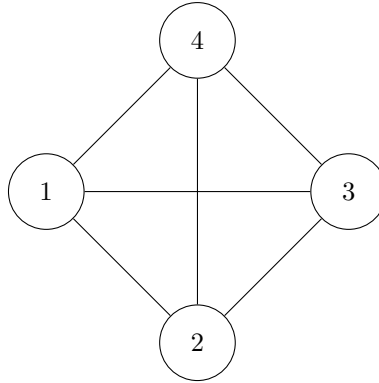
$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad (33)$$



The calculated dimensionless time is

$$\tilde{t} = 1.5701 \quad (34)$$

V. K_4



A. Analytical

The graph is represented by the Hamiltonian

$$H = -J \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad (35)$$

Using Mathematica, the eigenvalues are found to be

$$\lambda_1 = -3J \quad \lambda_2 = J \quad \lambda_3 = J \quad \lambda_4 = J \quad (36)$$

and their eigenvectors are

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad u_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad u_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad u_4 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (37)$$

B. Numerical

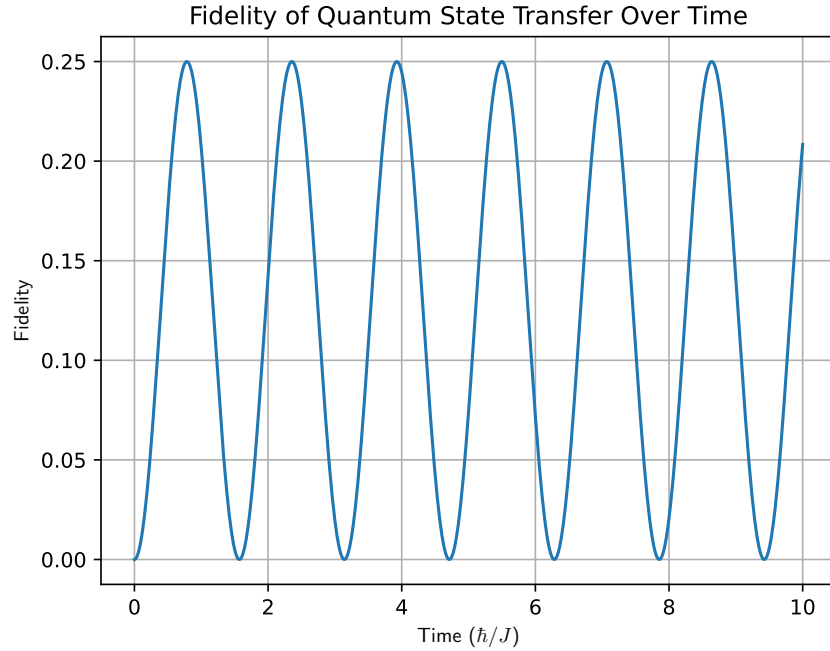
Assuming the initial state is localised on node 1, we have

$$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (38)$$

I am not sure about what the target state should be, so I tried different cases.

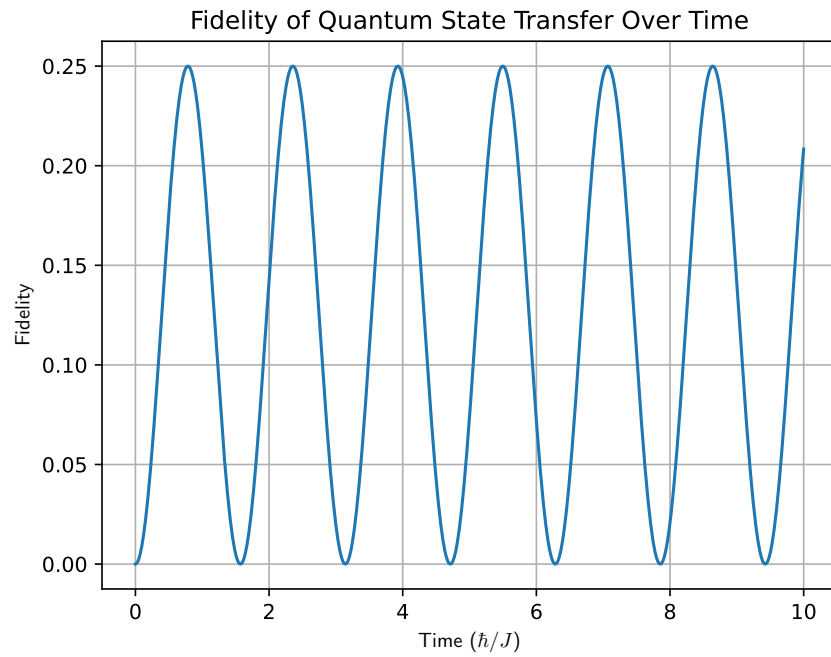
- When the target state is

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (39)$$



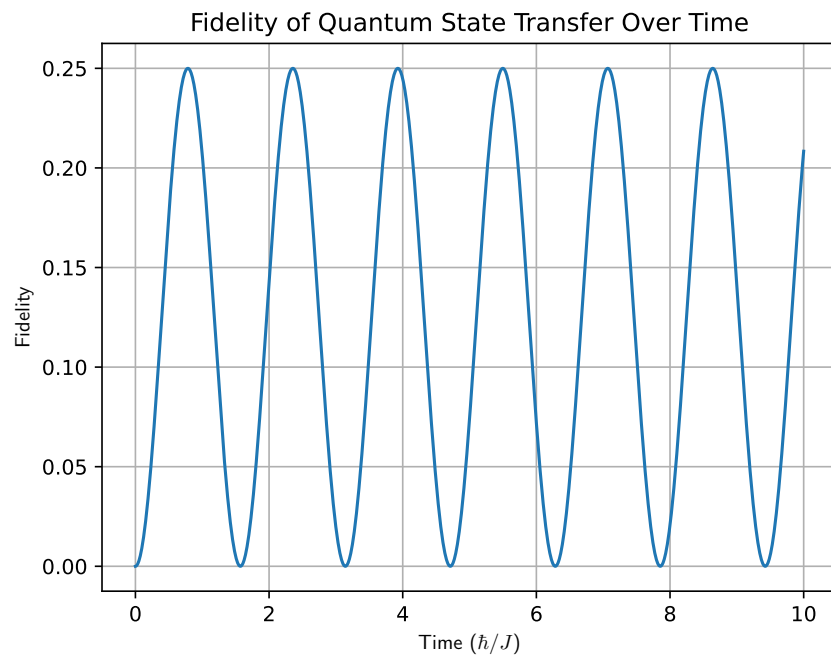
- When the target state is

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad (40)$$



- When the target state is

$$|\psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (41)$$



We can see that one step perfect quantum state transfer does not take place in the first 10 seconds.

VI. CONCLUSION

In summary, we conjecture that for a graph to allow one-step perfect quantum state transfer (PST), its eigenvalues must include a term proportional to $\sqrt{n-1}$, where n is the number of nodes in the graph. This condition on the eigenvalues suggests a specific energy spacing that facilitates the oscillation required for state transfer between nodes. For a particular class of graphs, such as the P -graphs that meet this eigenvalue condition, the time needed for one-step PST can be expressed as $\hat{t} = \pi/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue of the Hamiltonian matrix for the system. This relationship implies that the maximum energy separation in the system's spectrum dictates the speed of state transfer, reinforcing the connection between spectral properties and dynamical behavior in quantum networks. However, it is important to note that this hypothesis is based on a limited set of samples, and more comprehensive testing across a wider variety of graphs is needed to confirm its general validity. Further exploration may provide additional insights or reveal exceptions to this conjecture.

VII. CONDITIONS FOR PERFECT QUANTUM STATE TRANSFER AND PERIODICITY IN GRAPH

A. Necessary Conditions for PST

These conditions are required for PST to occur, meaning PST cannot happen without them.

1. Let $u_1, \dots, u_m$ be the eigenvectors in the spectral decomposition of $A(X)$ and let $\lambda_1, \dots, \lambda_m$ be the corresponding eigenvalues, then there is perfect quantum state transform from α to β at time τ if and only if there is constant γ such that

$$u_r|\alpha\rangle = \gamma \exp(-i\tau\lambda_r)u_r|\beta\rangle \quad (r = 1, \dots, m) \quad (42)$$

B. Sufficient Conditions for PST

These conditions ensure PST but are not strictly required.

1. If we have perfect quantum state transfer on X from α to β at time τ , then we must have perfect quantum state transfer from β to α at the same time τ .
2. Suppose X is a connected graph and X is periodic at α with minimum period σ . Then if there is perfect quantum state transfer from α to β , then there must be perfect quantum state transfer from β to α at time $\sigma/2$.
3. If we have perfect quantum state transfer from α to β in X at time τ , then at time τ we have perfect quantum state transfer in the d th Cartesian power of X between the d -tuples.

C. Derived Implications for PST

These are consequences of PST occurring or supplementary observations about its behavior.

1. If we have perfect quantum state transfer on X from α to β at time τ , then X must be periodic at α and β with period 2τ .
2. If we have perfect quantum state transfer in X from α to β and also from α to γ , then $\beta = \gamma$.

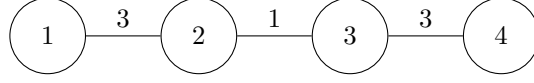
D. Sufficient conditions for Periodicity

These conditions ensure periodicity but are not strictly required.

1. The eigenvalues of X are integers
2. The eigenvalues of X are rational multipliers of $\sqrt{\Delta}$, for some square-free integer Δ .
3. If we have perfect quantum state transfer on X from α to β at time τ , then X must be periodic at α and β with period 2τ .

VIII. TWO-STEP PST OF P_4

A. Case 1



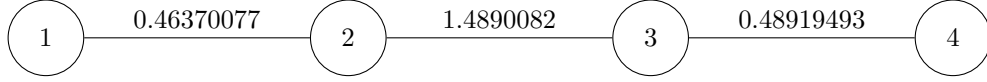
The initial Hamiltonian is

$$H = \begin{pmatrix} 0 & 3 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & 0 \end{pmatrix} \quad (43)$$

The states are

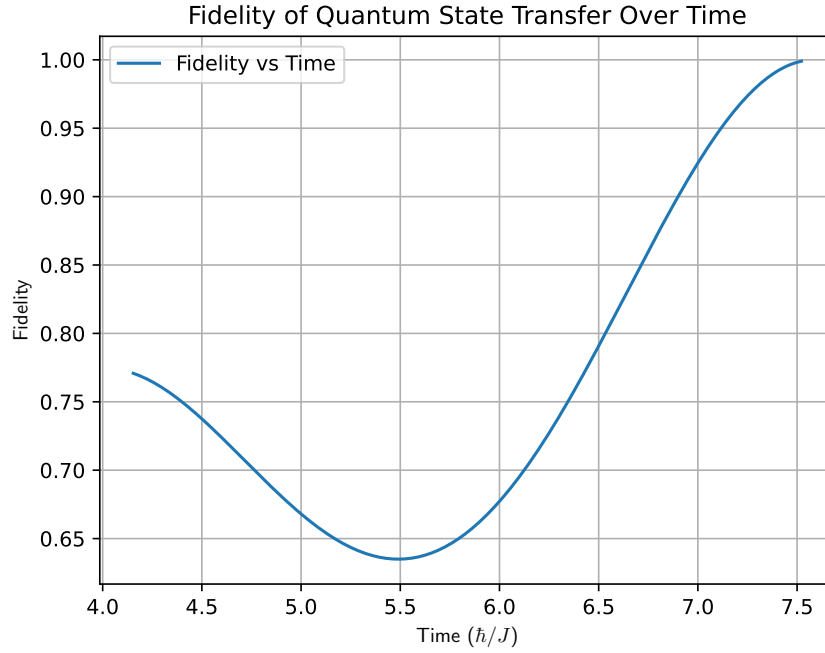
$$\psi_{initial} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \psi_{target} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (44)$$

For two-step perfect quantum state transfer to occur, I evolved the initial state until $\tilde{t} = 4.153943$. The fidelity at that time is calculated numerically to be 0.7708606. After that, I adjusted the edge weights:

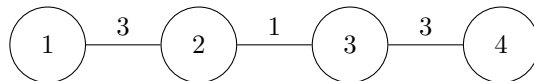


Then, I evolved the intermediate state, and a fidelity larger than 0.999 was calculated when

$$\tilde{t} = 7.5220 \quad (45)$$



B. Case 2



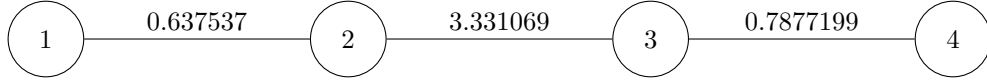
The initial Hamiltonian is

$$H = \begin{pmatrix} 0 & 3 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & 0 \end{pmatrix} \quad (46)$$

The states are

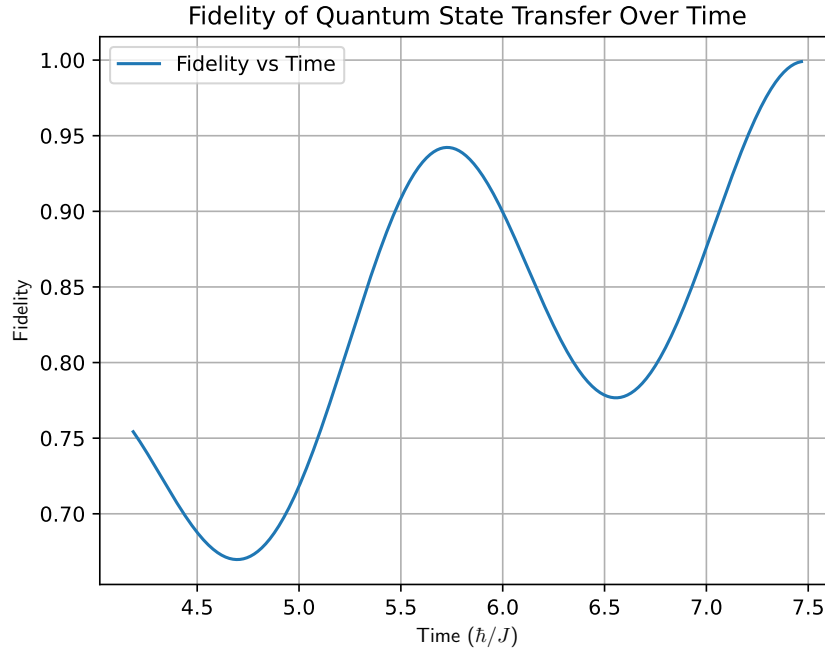
$$\psi_{initial} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \psi_{target} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (47)$$

For two-step perfect quantum state transfer to occur, I evolved the initial state until $\tilde{t} = 4.1858525$. The fidelity at that time is calculated numerically to be 0.7542758. After that, I adjusted the edge weights:



Then, I evolved the intermediate state, and a fidelity larger than 0.999 was calculated when

$$\tilde{t} = 7.4668 \quad (48)$$



IX. P₄: VALIDATOR

A. Time Evolution of Quantum States in Float64

1. Imports

Import the required libraries:

- `numpy` – for numerical operations
- `scipy.linalg.expm` – to compute matrix exponentials
- `matplotlib.pyplot` – for potential plotting (though not used directly)

2. Fidelity Calculation

1. Compute the time evolution operator:

$$U(t) = e^{-iHt}$$

2. Evolve the initial state:

$$\psi(t) = U(t)\psi_0$$

3. Calculate fidelity:

$$F = |\langle \psi_T | \psi(t) \rangle|^2$$

3. Hamiltonian Generation

Generate a 4×4 Hamiltonian for a path graph of four nodes:

$$H_{ij} = \begin{cases} w_1 & \text{if } (i, j) = (1, 2) \text{ or } (2, 1) \\ w_2 & \text{if } (i, j) = (2, 3) \text{ or } (3, 2) \\ w_3 & \text{if } (i, j) = (3, 4) \text{ or } (4, 3) \\ 0 & \text{otherwise} \end{cases}$$

4. Main Process

1. Receive user input for:

- Initial edge weights $[w_1, w_2, w_3]$
- Final edge weights $[w'_1, w'_2, w'_3]$
- Runtime before adjustment t_1
- Runtime after adjustment t_2

2. Generate the initial and final Hamiltonians:

$$H_1 = H(w), \quad H_2 = H(w')$$

3. Define the initial state (excitation at node 1):

$$\psi_0 = [1 \ 0 \ 0 \ 0]^T$$

4. Define the target state (excitation at node 3):

$$\psi_T = [0 \ 0 \ 1 \ 0]^T$$

5. Normalize:

$$\psi_0 = \frac{\psi_0}{\|\psi_0\|}, \quad \psi_T = \frac{\psi_T}{\|\psi_T\|}$$

6. Apply Hamiltonian H_1 for time t_1 :

$$F_1, \psi_1 = \text{calculate_fidelity}(H_1, \psi_0, \psi_T, t_1)$$

7. Apply H_2 to intermediate state ψ_1 for t_2 :

$$F_2, - = \text{calculate_fidelity}(H_2, \psi_1, \psi_T, t_2)$$

8. Output total fidelity:

$$F_{\text{total}} = F_2$$

5. Sample Output (Also as a Sanity Test)

```
Enter initial edge weights (space-separated, 3 values): 1 2 1
Enter final edge weights (space-separated, 3 values): 1 -2 1
Enter runtime before adjustment: 1.110720735
Enter runtime after adjustment: 1.110720735
```

Fidelity after full runtime: 0.9999999999999999333866185224906

X. P₄: OPTIMIZER

A. Symmetric Hamiltonians, Identical Runtime and Dependent Final Weights in Complex128

1. Imports

Import the necessary libraries:

- `numpy` – for numerical operations
- `tensorflow` – for automatic differentiation and optimization
- `matplotlib.pyplot` – for visualization
- `pandas` – for data storage and export
- `logging` – to suppress TensorFlow warnings

2. Hamiltonian Construction

Construct the Hamiltonian for the P₄ path graph using edge weights:

$$H = \begin{bmatrix} 0 & w_1 & 0 & 0 \\ w_1 & 0 & w_2 & 0 \\ 0 & w_2 & 0 & w_3 \\ 0 & 0 & w_3 & 0 \end{bmatrix}$$

For symmetric graphs, enforce $w_3 = w_1$.

3. Quantum State Evolution

State evolution over runtime t is governed by:

$$U(t) = e^{-iHt}$$

The fidelity between the evolved state $\psi(t)$ and target state ψ_T is:

$$F = |\langle \psi_T | U(t) | \psi_0 \rangle|^2$$

4. Optimization Process

1. Initialize edge weights $[w_1, w_2, w_1]$, runtimes t_1 and t_2 .
2. Define the initial and target states:

$$\psi_0 = [1 \ 0 \ 0 \ 0]^T, \quad \psi_T = [0 \ 0 \ 1 \ 0]^T$$

3. Evolve the state with Hamiltonian H_1 (initial weights) for time t_1 .
4. Use the intermediate state to evolve under Hamiltonian H_2 (final weights) for time t_2 .
5. Minimize the loss function:

$$\mathcal{L} = 1 - F$$

using the Adam optimizer.

6. Perturb the variables if the optimization stagnates over a window of 1000 steps.
7. Store solutions when fidelity reaches 1.0.

5. Stagnation Detection and Perturbation

If the fidelity change over the last N steps is below the threshold:

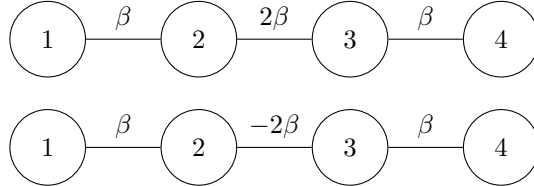
$$\max(F_{\text{recent}}) - \min(F_{\text{recent}}) < 0.0001$$

Perturb runtimes and edge weights by adding random noise.

6. Sample Output

Step 300: Loss = 0.000001, Fidelity = 1.000000,
 Initial Weights = [0.9543, 2.0301, 0.9543],
 Final Weights = [-0.9543, 2.0301, -0.9543],
 Runtime Before = 5.3492, Runtime After = 4.9817
 Solution Found at Step 300

XI. RESULTS



It was verified that in the configurations above, PST is possible from **node 1 and to node 3**. However,

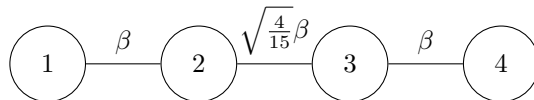
$$t = t' \neq \frac{\pi}{2\sqrt{2}\beta} \quad (49)$$

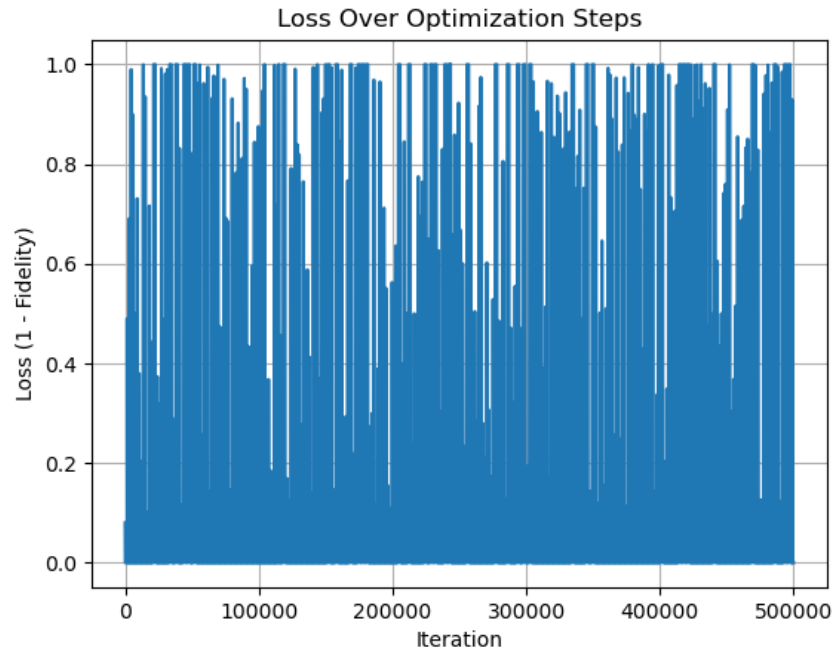
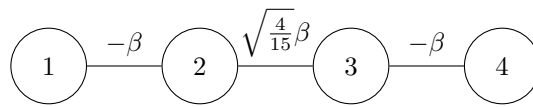
Interestingly,

$$t = t' = \frac{n\pi}{2\sqrt{2}\beta} \quad (50)$$

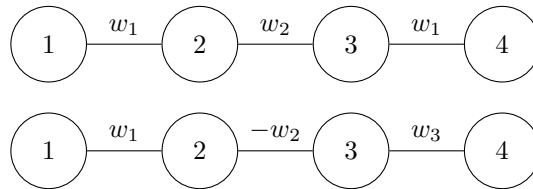
where n is a real positive integer.

However, another case demonstrated below, which is proven to exhibit PST from **node 1 to node 4**, cannot be found using the optimizer due to the fact that there are too many local minima in the loss value. The optimizer return no results most of the time.





I also found some solutions of PST from **node 1 to node 2** with the edge weights configurations below, but I have not attempted to work out the pattern yet.



One example is

```
Enter initial edge weights (space-separated, 3 values): 0.6626908221904778 -2.1850230509470592e-08
0.6626908221904778
Enter final edge weights (space-separated, 3 values): 0.6626908221904778 2.1850230509470592e-08
0.6626908221904778
Enter runtime before adjustment: 5.925826472411548
Enter runtime after adjustment: 5.925826472411548
```

```
Fidelity after full runtime: 1.00000000000000000000000000000000
```

This is the verification performed by the time evolution of quantum state code. The edge weight in the middle is pretty small, but it is still far from zero.

Here is another example:

```
Enter initial edge weights (space-separated, 3 values): 5.094830057973546 -6.170797557539656e-08
5.094830057973546
Enter final edge weights (space-separated, 3 values): 5.094830057973546 6.170797557539656e-08
5.094830057973546
Enter runtime before adjustment: 0.15415591008966015
```

Enter runtime after adjustment: 0.15415591008966015

Fidelity after full runtime: 1.000000000000000000000000000000

This one is a bit weird, I wonder what leads to a fidelity slightly larger than 1:

Enter initial edge weights (space-separated, 3 values): 5.885431761530667 8.174164514679293e-07
5.885431761530667

Enter final edge weights (space-separated, 3 values): 5.885431761530667 -8.174164514679293e-07
5.885431761530667

Enter runtime before adjustment: 6.005152856

Enter runtime after adjustment: 6.005152856

Fidelity after full runtime: 1.0000000000000007549516567451064

XII. QUESTIONS

1. Discrepancies between the fidelities output between the validator and optimizer, which is likely caused by the accuracy differences between `tf.linalg.expm` and `scipy.linalg.expm`, where the latter is more accurate. Would it be beneficial to code the Adam's algorithm from scratch?
2. Is 1 actually 1? Does a numerical fidelity of 1 guarantee PST?

XIII. P_4 : NODE 1 $\rightarrow$ NODE 4 PST

A. 0.999999999

Initial Edge Weights	Final Edge Weights	$t = t'$
[1.0, 2.2677864162435357, 1.0]	[-1.0, 2.2677864162435357, -1.0]	2.081701141
[1.0, 2.2677872585089562, 1.0]	[-1.0, 2.2677872585089562, -1.0]	2.0742356524309
[1.0, -2.267787257589282, 1.0]	[-1.0, -2.267787257589282, -1.0]	2.074246026
[1.0, -2.267786417033427, 1.0]	[-1.0, -2.267786417033427, -1.0]	2.08170109
[1.0, 0.5163978486429714, 1.0]	[-1.0, 0.5163978486429714, -1.0]	3.034013229
[1.0, -0.5163973327358463, 1.0]	[-1.0, -0.5163973327358463, -1.0]	3.044563193
[1.0, -0.5163978486459277, 1.0]	[-1.0, -0.5163978486459277, -1.0]	3.03401323
[1.0, 3.614820251255397, 1.0]	[-1.0, 3.614820251255397, -1.0]	3.04014415
[1.0, -3.614785401654898, 1.0]	[-1.0, -3.614785401654898, -1.0]	3.03887772
[1.0, -3.6147854023235837, 1.0]	[-1.0, -3.6147854023235837, -1.0]	3.038877381
[1.0, -4.587318653298133, 1.0]	[-1.0, -4.587318653298133, -1.0]	3.764012455
[1.0, -4.587316594892395, 1.0]	[-1.0, -4.587316594892395, -1.0]	11.30253673
[1.0, 4.587345143460048, 1.0]	[1.0, 4.587345143460048, 1.0]	3.767587833
[1.0, 5.388161262251575, 1.0]	[-1.0, 5.388161262251575, -1.0]	4.370490101
[1.0, -5.388204386530768, 1.0]	[-1.0, -5.388204386530768, -1.0]	4.371473889
[1.0, -1.6012814463605203, 1.0]	[-1.0, -1.6012814463605203, -1.0]	4.909252167
[1.0, -1.6012814463922922, 1.0]	[-1.0, -1.6012814463922922, -1.0]	4.909252043
[1.0, -1.6012816295409682, 1.0]	[-1.0, -1.6012816295409682, -1.0]	4.9003682
[1.0, 6.084872739875749, 1.0]	[-1.0, 6.084872739875749, -1.0]	4.902530395
[1.0, -6.084866925228957, 1.0]	[-1.0, -6.084866925228957, -1.0]	4.907089391
[1.0, -6.084866943257897, 1.0]	[-1.0, -6.084866943257897, -1.0]	4.907089902
[1.0, -6.709793237392034, 1.0]	[-1.0, -6.709793237392034, -1.0]	5.382247619
[1.0, -6.709793238127481, 1.0]	[-1.0, -6.709793238127481, -1.0]	5.382247074
[1.0, -6.709785918264255, 1.0]	[-1.0, -6.709785918264255, -1.0]	5.386588214
[1.0, 0.809039874164738, 1.0]	[-1.0, 0.809039874164738, -1.0]	5.818418956
[1.0, 0.8090386679103602, 1.0]	[-1.0, 0.8090386679103602, -1.0]	5.824430125
[1.0, -0.8090398741444761, 1.0]	[-1.0, -0.8090398741444761, -1.0]	5.818420127
[1.0, -7.281369869968289, 1.0]	[-1.0, -7.281369869968289, -1.0]	5.822991142
[1.0, -7.2813540562574275, 1.0]	[1.0, -7.2813540562574275, 1.0]	5.826753211
[1.0, -7.2813541158043575, 1.0]	[-1.0, -7.2813541158043575, -1.0]	5.826752993
[1.0, -0.2519763555275555, 1.0]	[-1.0, -0.2519763555275555, -1.0]	6.222707537
[1.0, 7.81127097891731, 1.0]	[-1.0, 7.81127097891731, -1.0]	6.231891388
[1.0, -7.811218900466319, 1.0]	[-1.0, -7.811218900466319, -1.0]	6.235755067
[1.0, -7.811260571812177, 1.0]	[-1.0, -7.811260571812177, -1.0]	6.235917089
[1.0, -8.307465564796429, 1.0]	[-1.0, -8.307465564796429, -1.0]	6.619834791
[1.0, -8.307477689698592, 1.0]	[-1.0, -8.307477689698592, -1.0]	6.615930995
[1.0, -8.775678756720122, 1.0]	[1.0, -8.775678756720122, 1.0]	6.982671079
[1.0, -8.775692480914591, 1.0]	[-1.0, -8.775692480914591, -1.0]	6.978872379
[1.0, -8.775692507093504, 1.0]	[-1.0, -8.775692507093504, -1.0]	6.978872542
[1.0, -2.7874927851808353, 1.0]	[-1.0, -2.7874927851808353, -1.0]	7.322339243
[1.0, 2.7874923958166526, 1.0]	[-1.0, 2.7874923958166526, -1.0]	7.329073396
[1.0, -9.220175551480535, 1.0]	[-1.0, -9.220175551480535, -1.0]	7.323852643
[1.0, -9.220160041813735, 1.0]	[-1.0, -9.220160041813735, -1.0]	7.327560054
[1.0, -9.644230191589418, 1.0]	[-1.0, -9.644230191589418, -1.0]	7.653361319

[1.0, -1.4363696430269532, 1.0]	[-1.0, -1.4363696430269532, -1.0]	7.659804294
[1.0, -1.4363701167830498, 1.0]	[-1.0, -1.4363701167830498, -1.0]	7.650430304
[1.0, 10.050348592504676, 1.0]	[-1.0, 10.050348592504676, -1.0]	7.97269521
[1.0, -10.440748415133244, 1.0]	[-1.0, -10.440748415133244, -1.0]	8.272939815
[1.0, -3.227133528574531, 1.0]	[-1.0, -3.227133528574531, -1.0]	8.274232835
[1.0, -3.2271369323243624, 1.0]	[-1.0, -3.2271369323243624, -1.0]	8.277812482
[1.0, 0.16724843366067466, 1.0]	[-1.0, 0.16724843366067466, -1.0]	9.378249982
[1.0, 3.9652583074360974, 1.0]	[-1.0, 3.9652583074360974, -1.0]	9.900671203
[1.0, -3.96525830774512, 1.0]	[-1.0, -3.96525830774512, -1.0]	9.900670429
[1.0, -4.287482925652273, 1.0]	[-1.0, -4.287482925652273, -1.0]	10.6273849

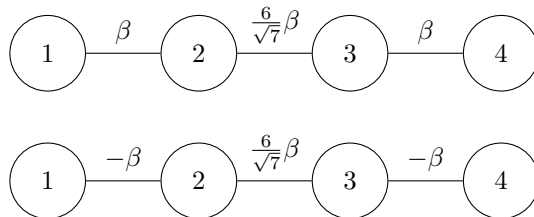
B. 0.9999999999999999

Initial Edge Weights	Final Edge Weights	$t = t'$
[1.0, -2.267786840422201, 1.0]	[-1.0, -2.267786840422201, -1.0]	2.077304562
[1.0, -2.267786840422201, 1.0]	[-1.0, -2.267786840422201, -1.0]	2.077304562
[1.0, -2.267786840422201, 1.0]	[-1.0, -2.267786840422201, -1.0]	2.077304562
[1.0, 3.6147844511476226, 1.0]	[-1.0, 3.6147844511476226, -1.0]	3.042359822
[1.0, 3.6147844511476226, 1.0]	[-1.0, 3.6147844511476226, -1.0]	3.042359822
[1.0, 4.587317100652229, 1.0]	[-1.0, 4.587317100652229, -1.0]	3.767103882
[1.0, 0.5163977791040121, 1.0]	[-1.0, 0.5163977791040121, -1.0]	3.043224841
[1.0, 0.5163979082935659, 1.0]	[-1.0, 0.5163979082935659, -1.0]	3.041398719
[1.0, -5.388159048452486, 1.0]	[-1.0, -5.388159048452486, -1.0]	4.373342537
[1.0, -5.388159073147655, 1.0]	[-1.0, -5.388159073147655, -1.0]	4.37248137
[1.0, 1.6012815385662786, 1.0]	[-1.0, 1.6012815385662786, -1.0]	4.904019911
[1.0, 0.8090399266807671, 1.0]	[-1.0, 0.8090399266807671, -1.0]	5.823707582
[1.0, 0.8090398347456658, 1.0]	[-1.0, 0.8090398347456658, -1.0]	5.825780094
[1.0, 0.8090398347456658, 1.0]	[-1.0, 0.8090398347456658, -1.0]	5.825780094
[1.0, 6.709789588696451, 1.0]	[-1.0, 6.709789588696451, -1.0]	5.384804545
[1.0, -0.2519763151128192, 1.0]	[-1.0, -0.2519763151128192, -1.0]	6.235895717

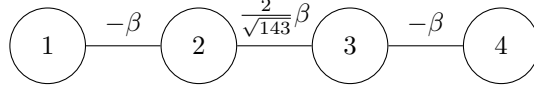
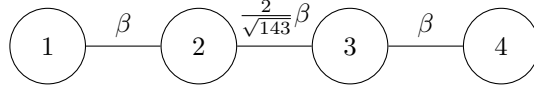
C. 0.9999999999999999

Initial Edge Weights	Final Edge Weights	$t = t'$
[1.0, -2.2677868380760757, 1.0]	[-1.0, -2.2677868380760757, -1.0]	2.077831382
[1.0, -2.2677868380760757, 1.0]	[-1.0, -2.2677868380760757, -1.0]	2.077831382
[1.0, -7.281358514018219, 1.0]	[-1.0, -7.281358514018219, -1.0]	5.824775131
[1.0, 0.1672484025560938, 1.0]	[-1.0, 0.1672484025560938, -1.0]	9.392512825

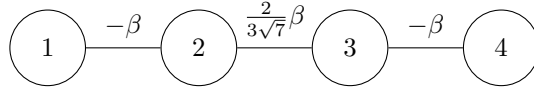
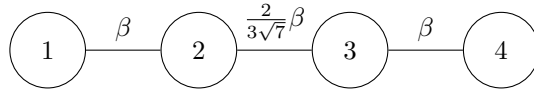
D. Exact Solutions



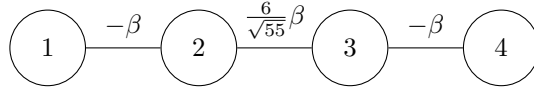
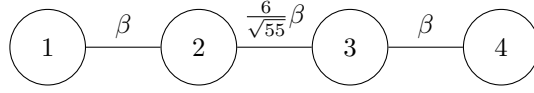
$$t = t' = \frac{\sqrt{7}}{4\beta} \pi$$



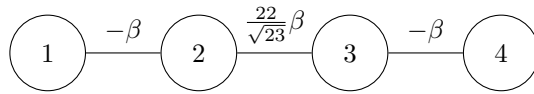
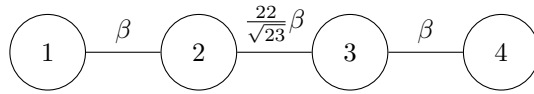
$$t = t' = \frac{\sqrt{143}}{4\beta} \pi$$



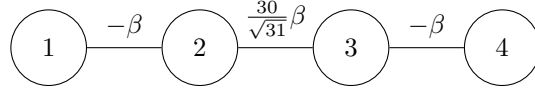
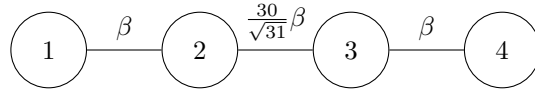
$$t = t' = \frac{3\sqrt{7}}{4\beta} \pi$$



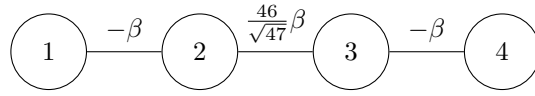
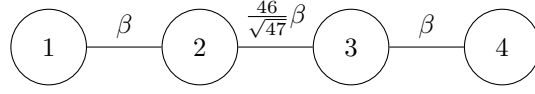
$$t = t' = \frac{\sqrt{55}}{4\beta} \pi$$



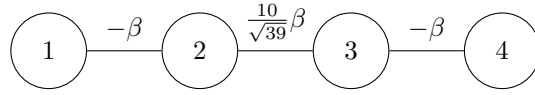
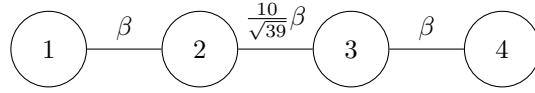
$$t = t' = \frac{\sqrt{23}}{4\beta} \pi$$



$$t = t' = \frac{\sqrt{31}}{4\beta} \pi$$

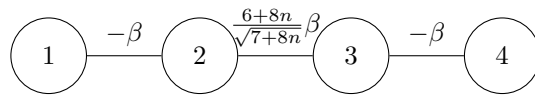
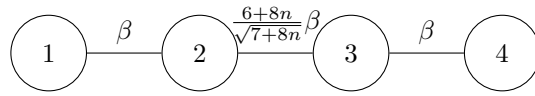


$$t = t' = \frac{\sqrt{47}}{4\beta} \pi$$



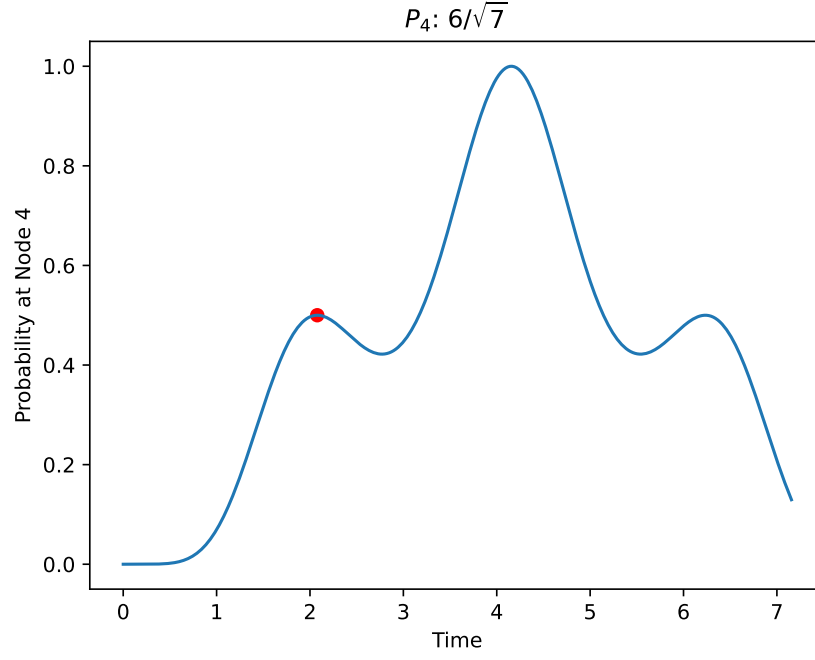
$$t = t' = \frac{\sqrt{39}}{4\beta} \pi$$

E. Possible Relations

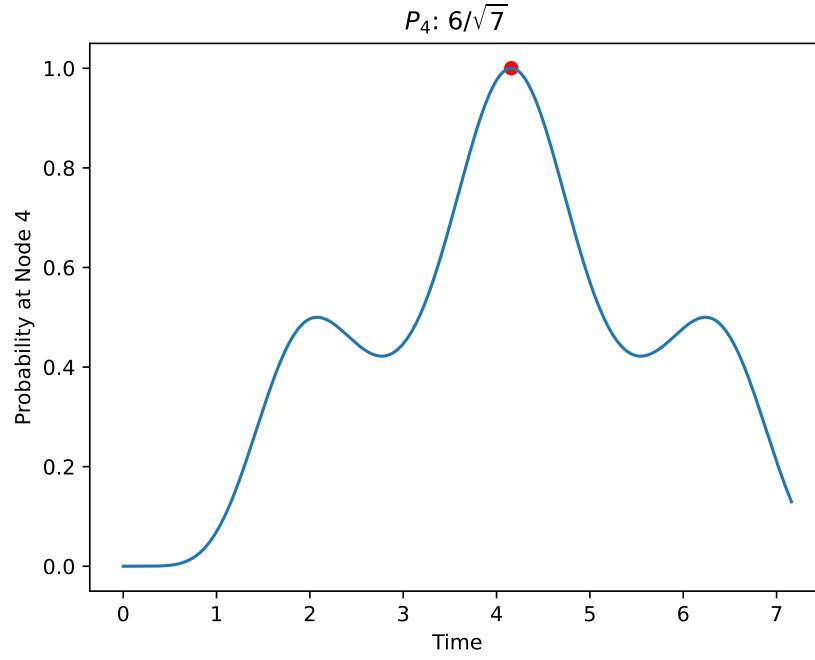


$$t = t' = \frac{\sqrt{7+8n}}{4\beta} \pi, \quad (n \in \mathbb{Z})$$

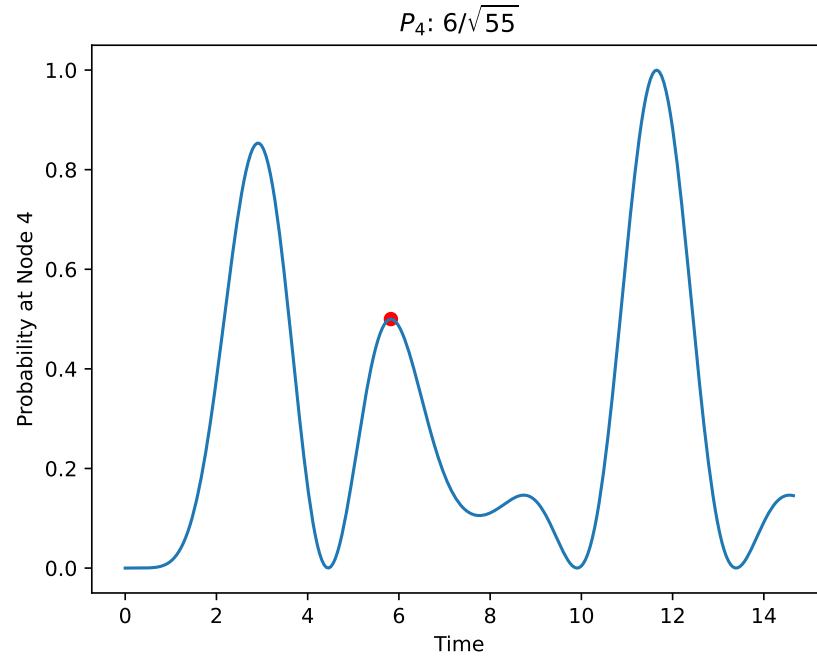
F. $t = t'$ and $t \neq t'$



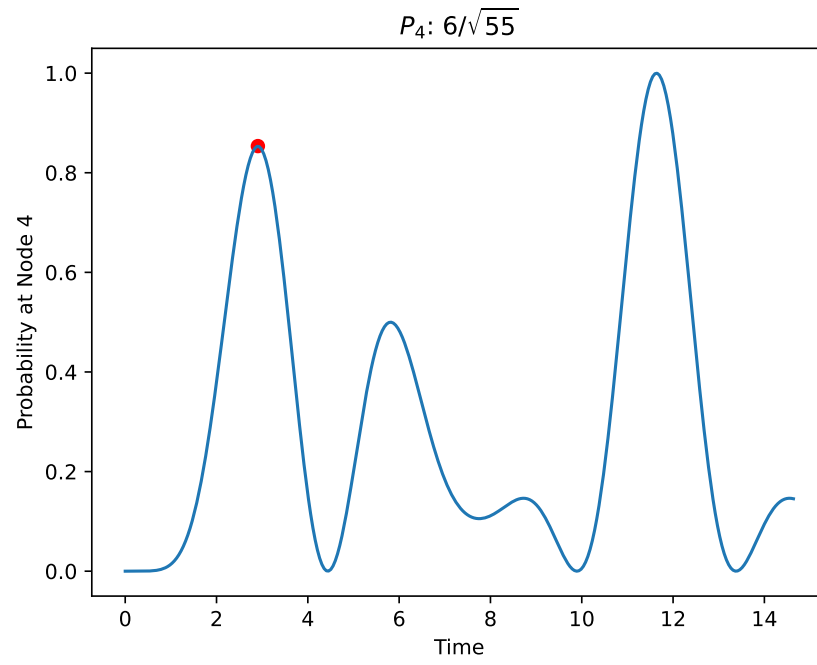
$$t = t' = \frac{\sqrt{7}}{4}\pi$$



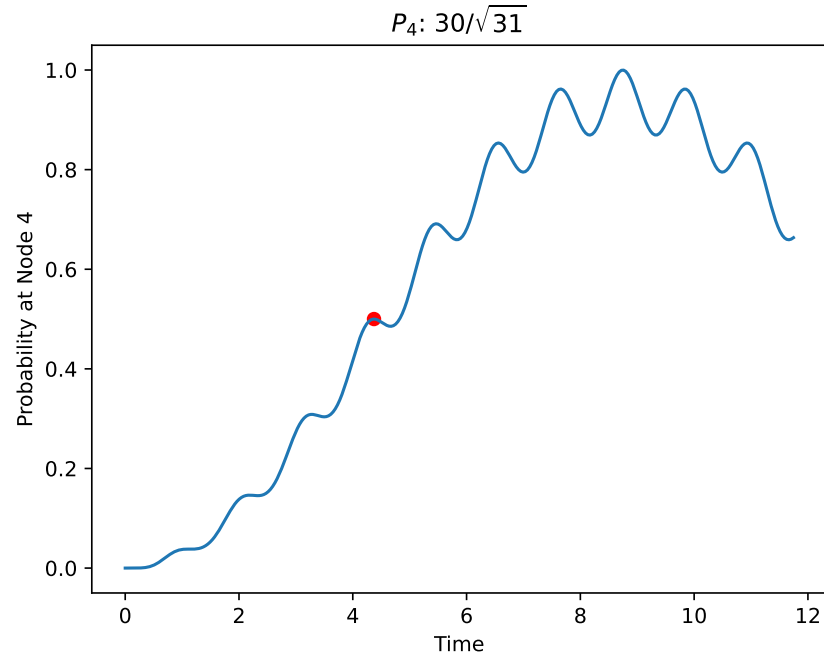
$$t = 4.157761123, \quad t' = 0.001787291$$



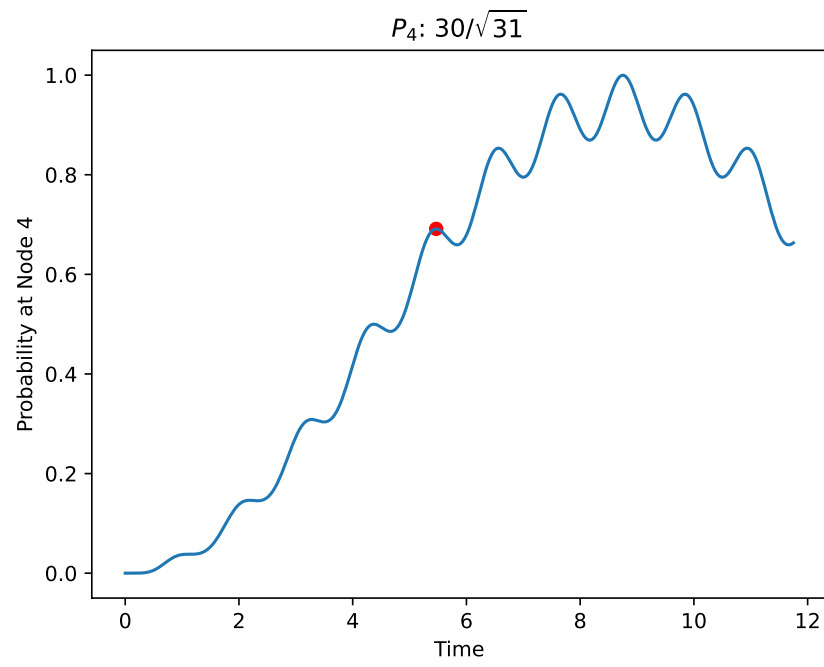
$$t = t' = \frac{\sqrt{55}}{4} \pi$$



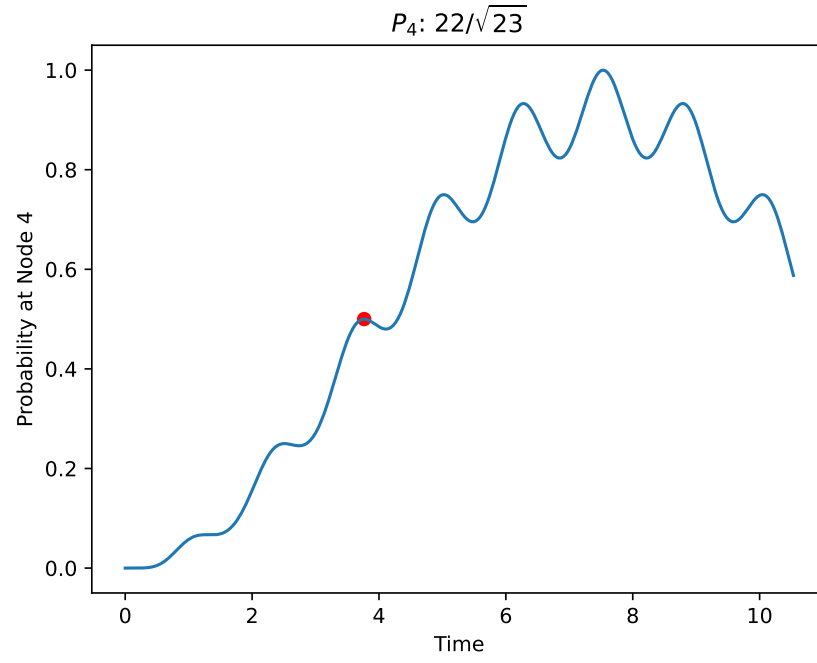
$$t = 2.906083132, \quad t' = 8.730751769$$



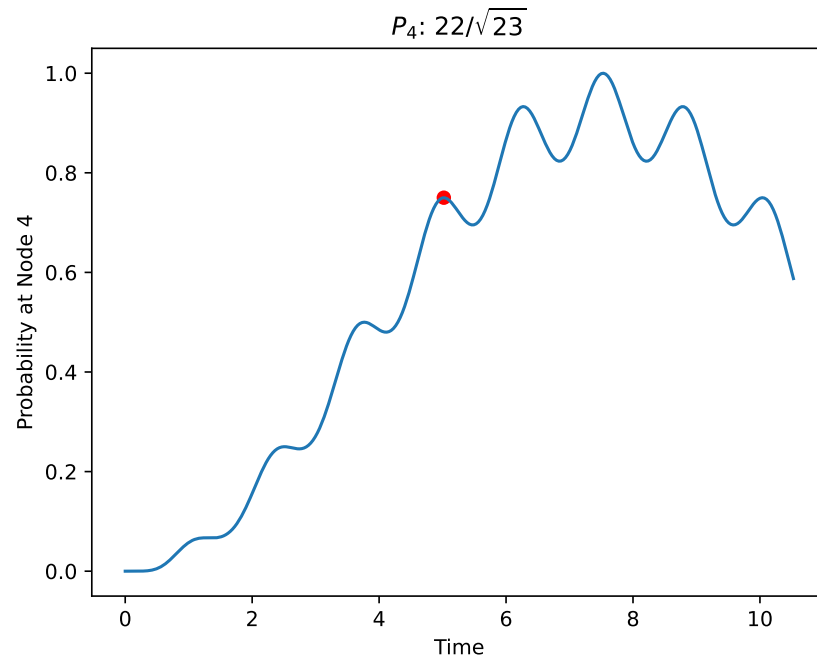
$$t = t' = \frac{\sqrt{31}}{4}\pi$$



$$t = 5.468562326, \quad t' = 3.282105682$$

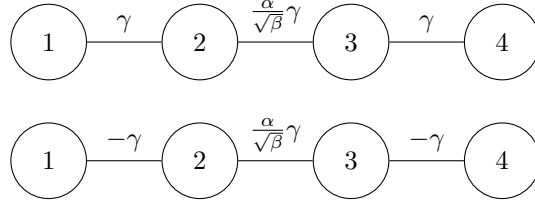


$$t = t' = \frac{\sqrt{23}}{4}\pi$$



$$t = 5.019558545, \quad t' = 2.50846772$$

G. The Golden Rule?



$$\beta = 7 + 8n = (5 + p)(3 - p + 8m),$$

$$\alpha = 2 + 2p - 8m,$$

$$\text{where } \begin{cases} n = 0, 1, 2, \dots, \\ m = 0, 1, 2, \dots, n. \end{cases}$$

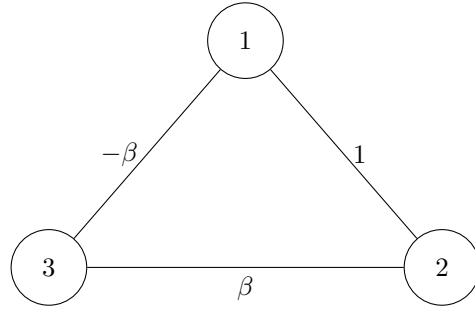
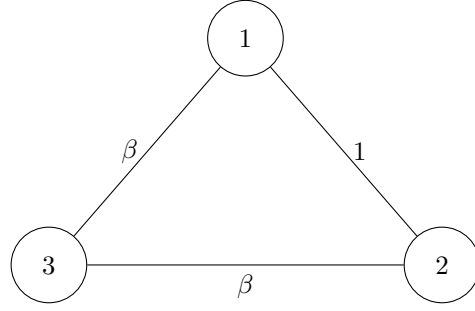
If $p \in \mathbb{Z}$ and $\alpha/\sqrt{\beta}$ is in its simplest form $\alpha'/\sqrt{\beta'}$ where $\alpha', \beta' \in \mathbb{Z}$, then the shortest time needed for two-step PST is

$$t = t' = \frac{\sqrt{\beta'}\pi}{4\gamma}.$$

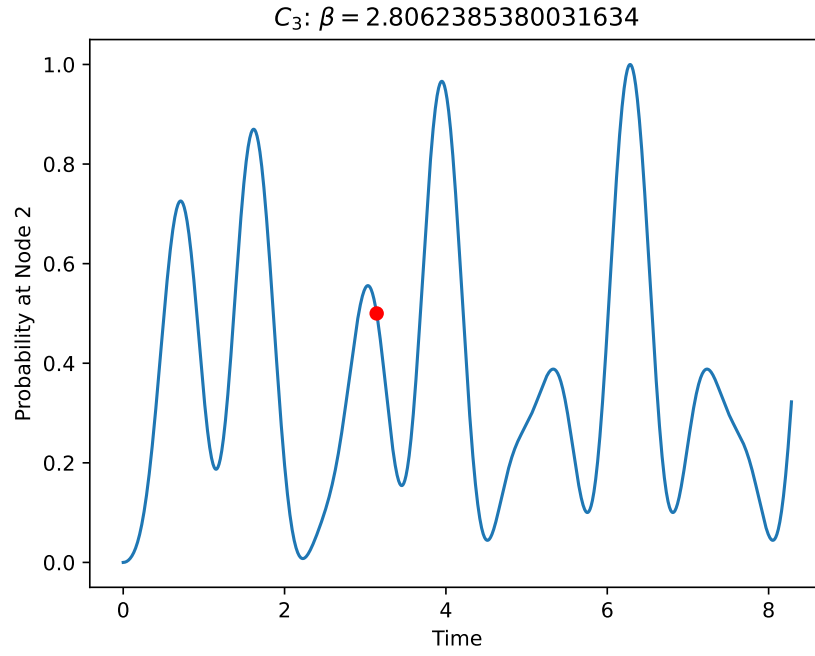
The shortest time needed for one-step PST is

$$t = t' = \frac{\sqrt{\beta'}\pi}{2\gamma}.$$

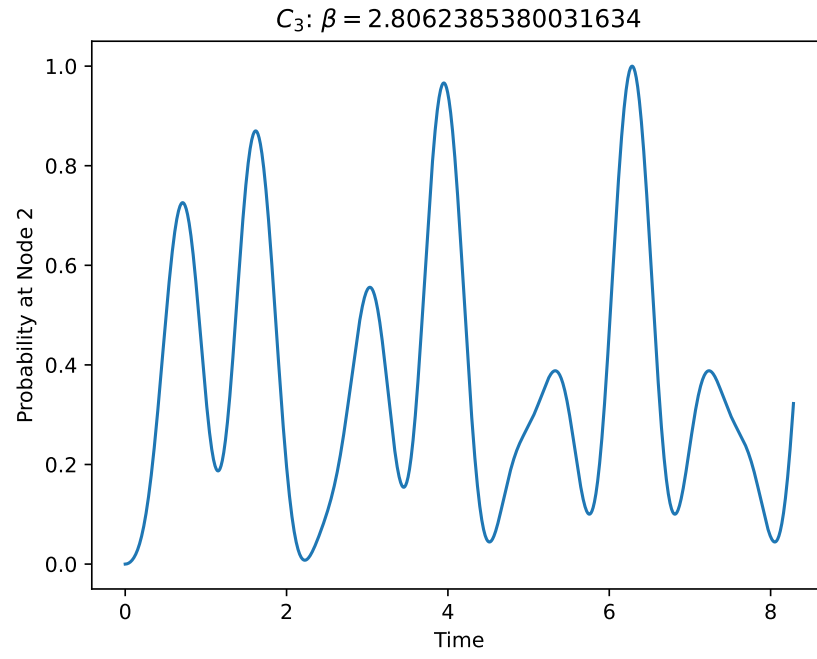
XIV. C_3 NODE 1 $\rightarrow$ NODE 2 PST



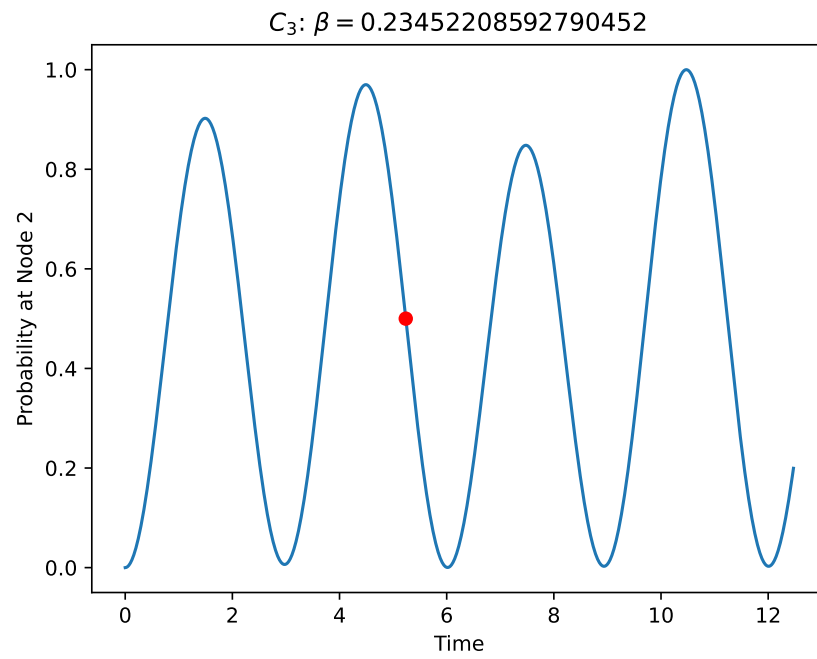
A. Is Two-Step Necessary?



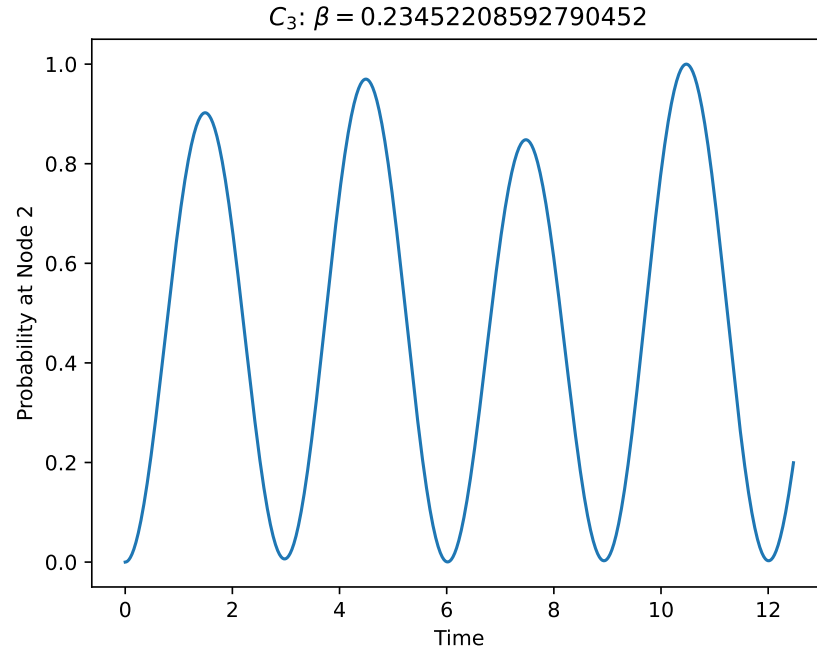
$$t = t' = \pi$$



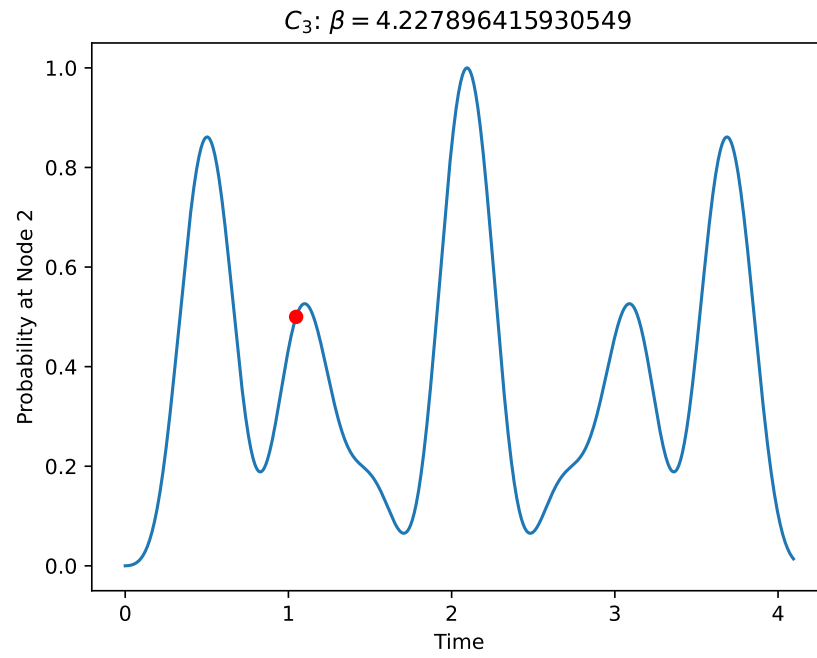
$$t = 2\pi$$



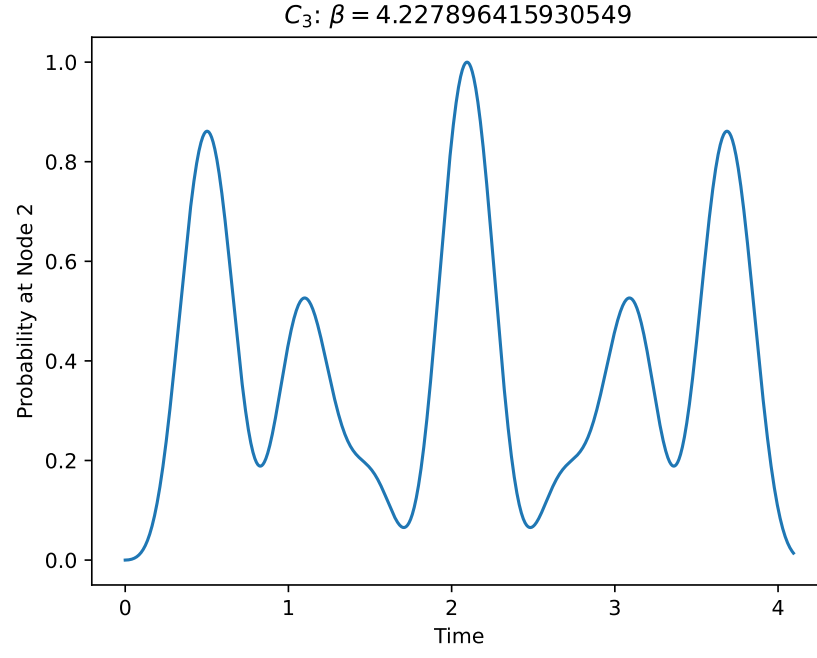
$$t = t' = \frac{5\pi}{3}$$



$$t = \frac{10\pi}{3}$$



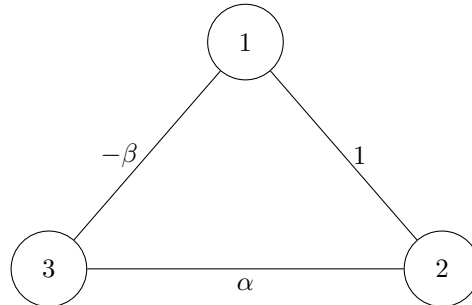
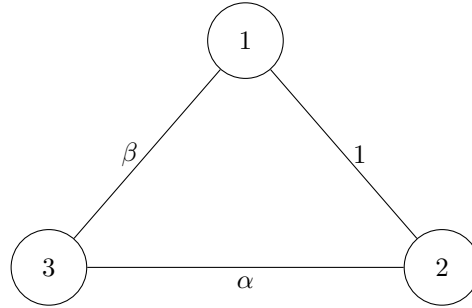
$$t = t' = \frac{\pi}{3}$$



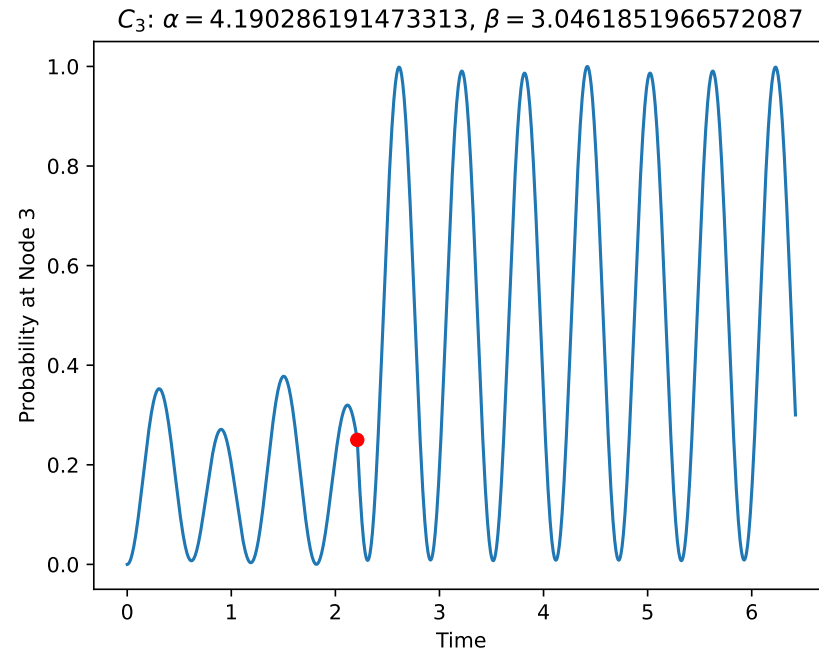
$$t = \frac{2\pi}{3}$$

The second step appears to be unnecessary in this family.

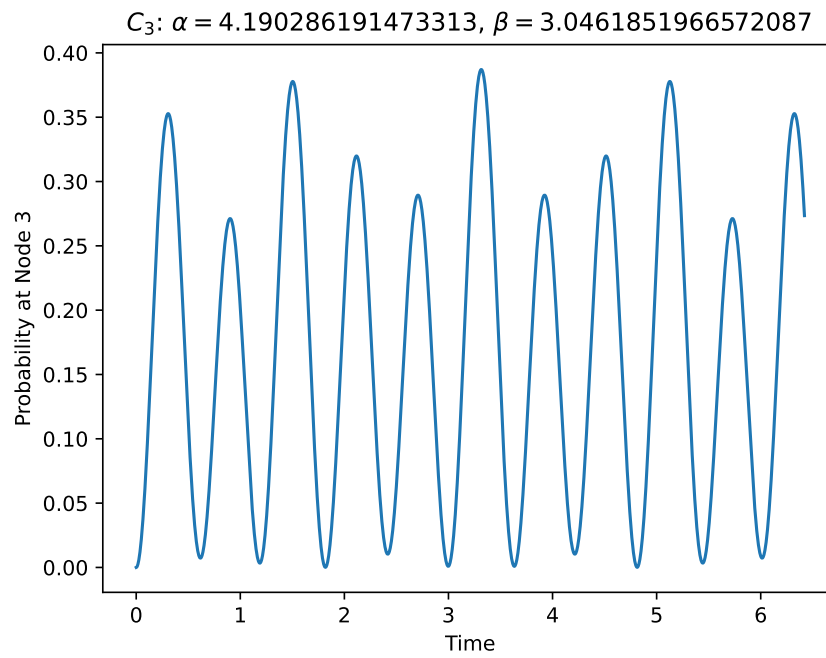
XV. C_3 NODE 1 $\rightarrow$ NODE 3 PST



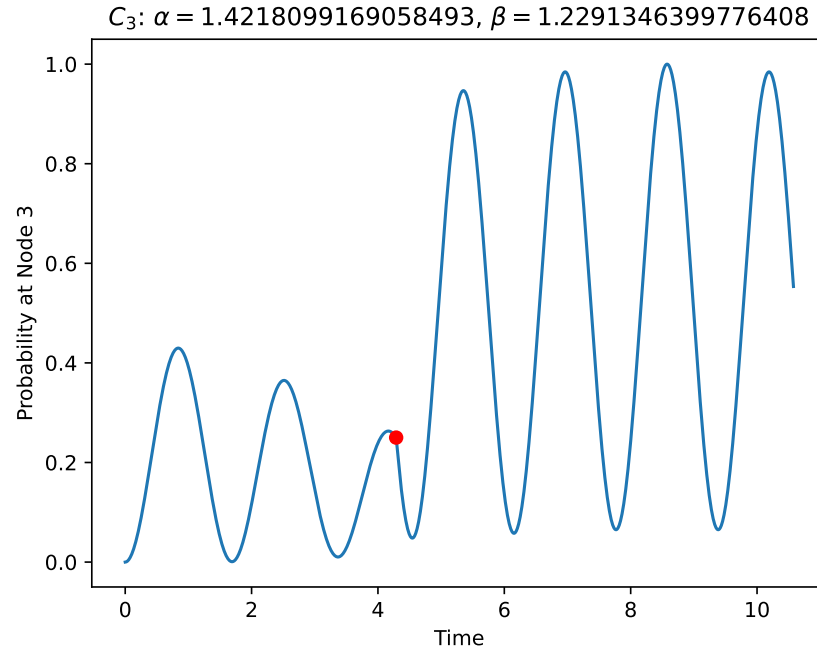
A. Two-Step isn't Useless!!!



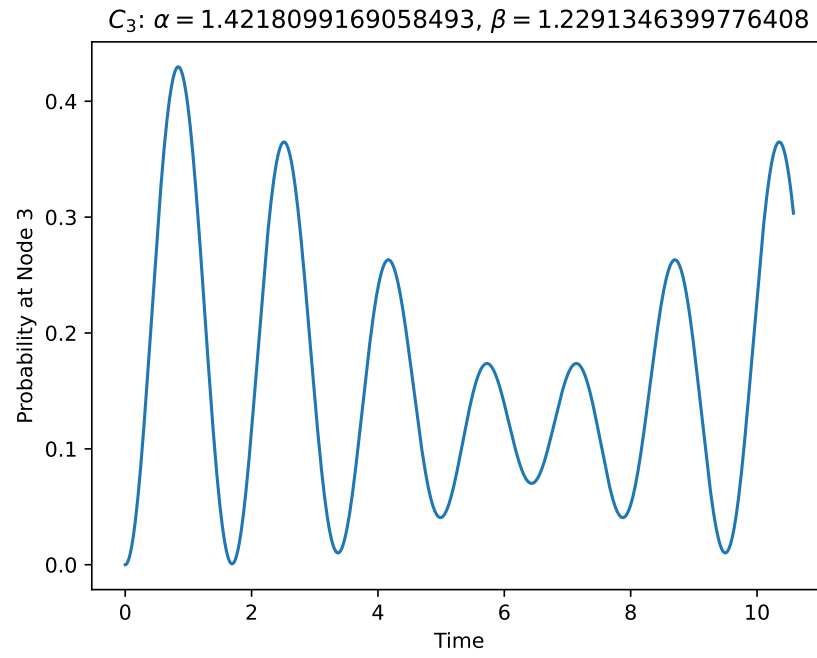
$$t = t' = 2.210153548$$



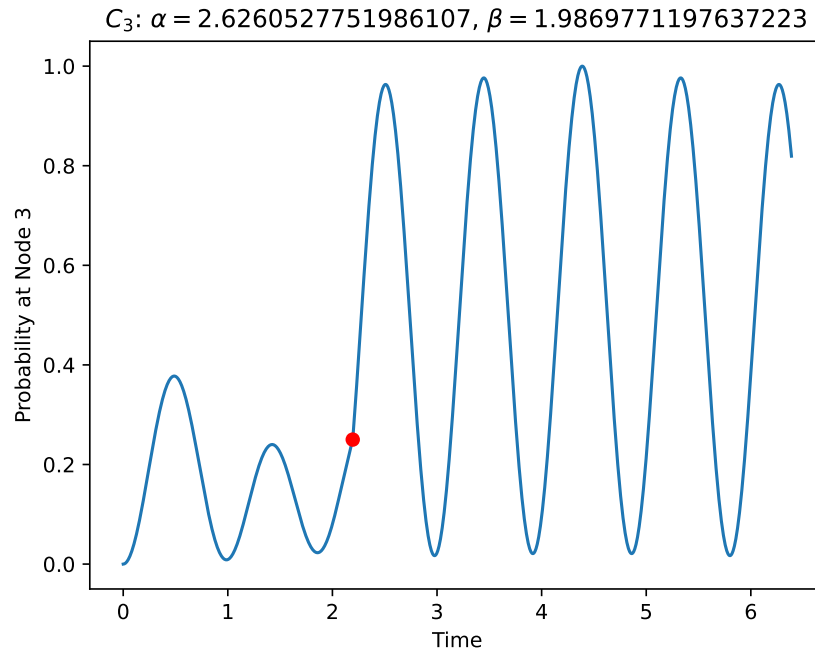
$$t = 4.420307096$$



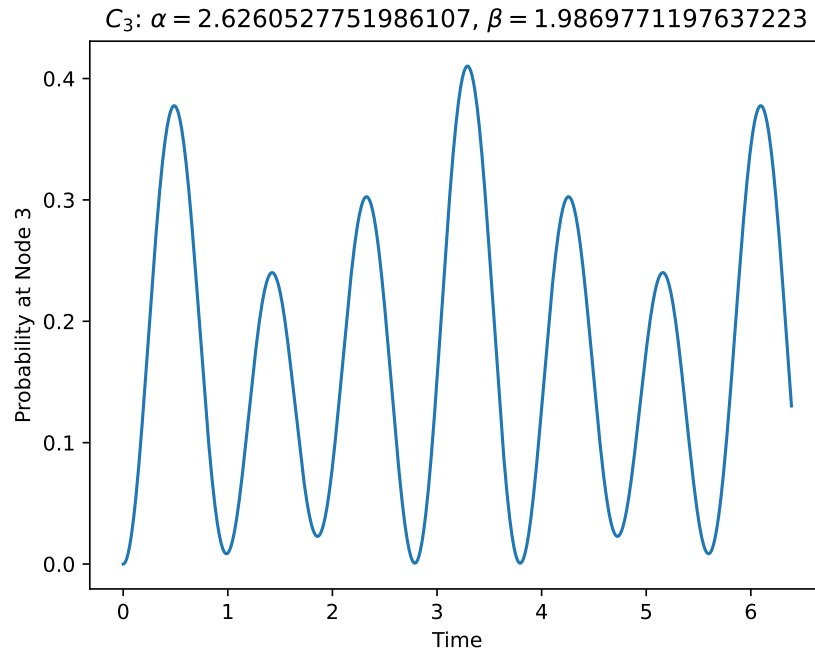
$$t = t' = 4.288203328$$



$$t = 8.576406656$$



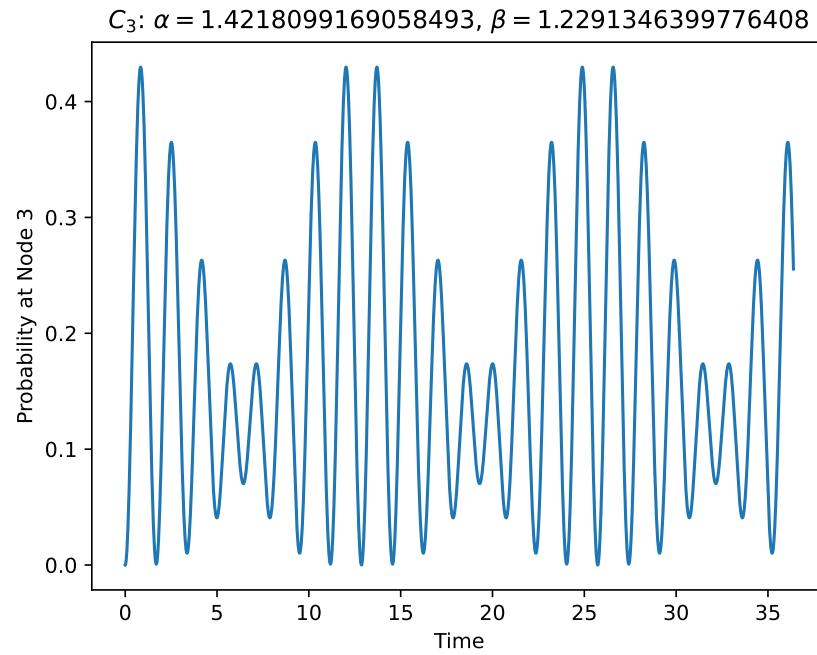
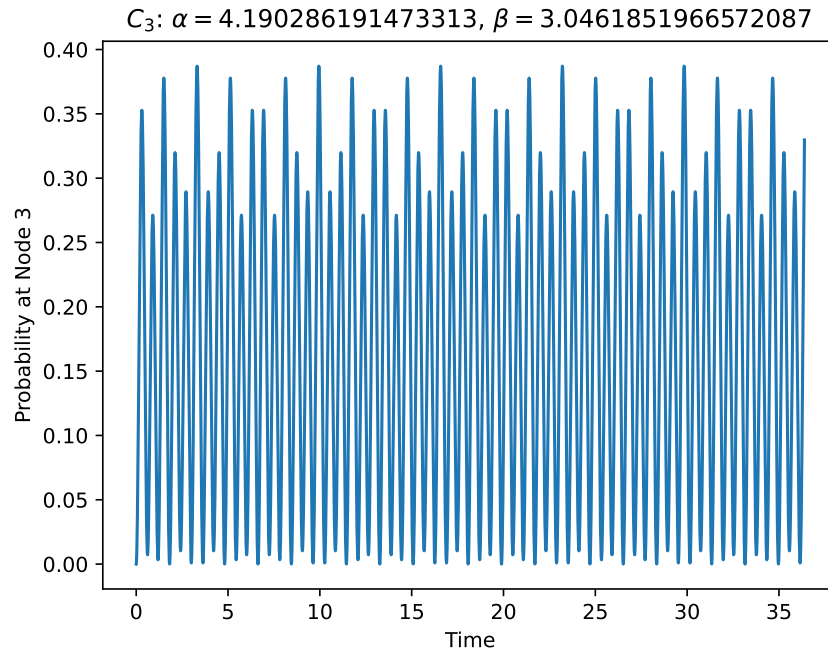
$$t = t' = 2.19420324$$

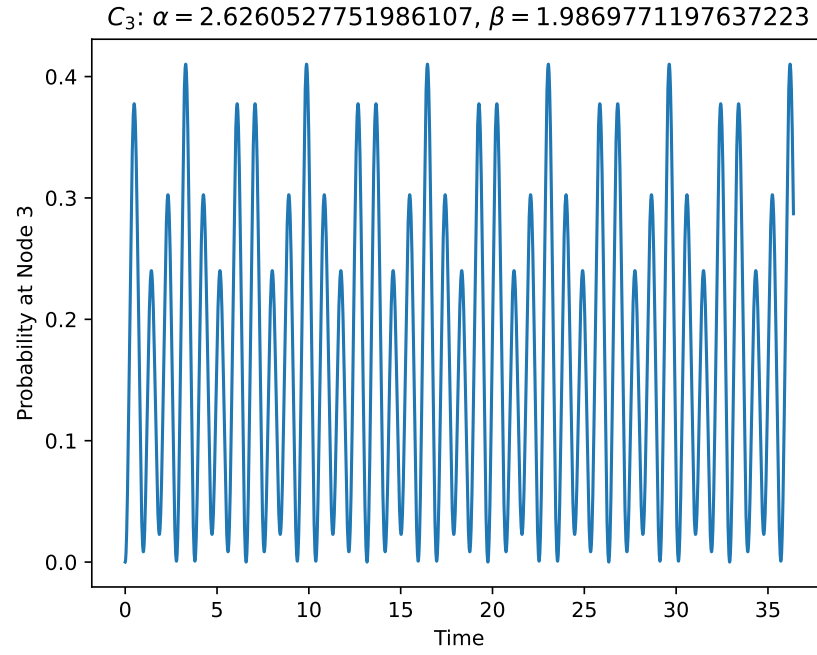


$$t = 4.38840648$$

When I was working on P_4 , I got many results on asymmetric edge weights at first, but I thought they were all PGST and ignored them. Now thinking back, two-step PST might be the only way to achieve perfect fidelity on asymmetric graphs.

B. Checking for Periodicity in One-Step PST





I think it is safe to say that without the second step, the fidelity never reach 1, not even close.

XVI. APPENDIX

A. Proof of Eq. 6

We know that

$$e^x = \sum_{n=0}^{\infty} x^n / n! \quad (51)$$

Therefore, by $\langle u_n | u_m \rangle = \delta_{nm}$, $e^{-iHt/\hbar}$ can be rewritten as

$$\begin{aligned} e^{-\frac{iHt}{\hbar}} &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{iHt}{\hbar} \right)^n \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{it}{\hbar} \sum_m \lambda_m |u_m\rangle \langle u_m| \right)^n \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_m \left(-\frac{i\lambda_m t}{\hbar} \right)^n |u_m\rangle \langle u_m| \\ &= \sum_m \left(\sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i\lambda_m t}{\hbar} \right)^n \right) |u_m\rangle \langle u_m| \\ &= \sum_m e^{-\frac{i\lambda_m t}{\hbar}} |u_m\rangle \langle u_m| \\ &= \sum_n e^{-\frac{i\lambda_n t}{\hbar}} |u_n\rangle \langle u_n| \end{aligned} \quad (52)$$

B. Equation to be Solved

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = e^{-iH't/\hbar} \left[e^{-iHt/\hbar} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right] \quad (53)$$

where

$$H = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 6/\sqrt{7} & 0 \\ 0 & 6/\sqrt{7} & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (54)$$

$$H = \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 6/\sqrt{7} & 0 \\ 0 & 6/\sqrt{7} & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad (55)$$